BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024 • Paper Code: 6195 / 6196

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q.PAPER - I (Objective Type)

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 6195

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Rank of the matrix [1; 0; -1] is :

(A) 0 (B) 1

(C) 2 (D) 3

Q2. The fraction $(x+1)/(x^2+2)$ is :

(A) Improper fraction (B) Proper fraction

(C) Identity (D) Mixed

Q3. The multiplicative inverse of (1 , 0) is :

(A) (1 , 0) (B) (0 , 1)

(C) (- 1 , 0) (D) (0 , - 1)

Q4. The roots of $2x^2 - 7x + 3 = 0$, are :

(A) Equal (B) Complex

(C) Irrational (D) Rational

Q5. The value of $(-i)^9$ is :

(A) - 1 (B) 1

(C) i (D) - i

Q6. If A is a square matrix of order 3 and |A| = 2 , then |2A| = :

(A) 16 (B) 8

(C) 6 (D) 2

Q7. The number of elements of the power set of A = { a, { b , c } } are :

(A) 2 (B) 4

(C) 6 (D) 8

Q8. If $A \subseteq B$, then :

(A) $A \cup B = A$ (B) $A \cap B = B$

(C) $B \cup A = A$ (D) $A \cup B = B$

Q9. If ω is a cube root of unity, then value of $(1+\omega-\omega^2)^3$ is :

(A) 8ω (B) $8\omega^2$

(C) - 8 (D) 8

Q10. The converse of $\sim p \rightarrow q$ is :

(A) $p \rightarrow q$ (B) $p \rightarrow \sim q$

(C) $\sim q \rightarrow p$ (D) $q \rightarrow \sim p$

Q11. cos 2θ =:

(A) $1 - \sin^2\theta$ (B) $1 - 2\sin\theta$

(C) $1 - 2\sin^2\theta$ (D) $2\sin^2\theta - 1$

Q12. The G.M. between 1/a and 1/b is :

(A) $\pm\sqrt{ab}$ (B) $\pm 1/ab$

(C) $\pm \sqrt{(1/ab)}$ (D) ab

Q13. If cos x = -√3/2 , then the reference angle is :

(A) π/3 (B) π/6

(C) -π/3 (D) -π/6

Q14. If sin θ < 0 and cot θ > 0 , then θ lies in quadrant :

(A) IV (B) III

(C) II (D) I

Q15. The value of $\sin^{-1}(\cos \pi/6)$ is equal to :

(A) π/3 (B) π/6

(C) π/2 (D) 3π/2

Q16. The relation between A, G, H is :

(A) $G^2 = AH$ (B) $H^2 = AG$

(C) $A^2 = HG$ (D) $A > G < H$

Q17. The number of terms in the expansion of $(a+x)^n$ is :

(A) n - 1 (B) n

(C) n + 2 (D) n + 1

Q18. $\sqrt{(s(s-c)/ab)}$ = :

(A) cos α/2 (B) sin α/2

(C) cos γ/2 (D) sin γ/2

Q19. A die is thrown, what is the probability to get 3 dots :

(A) 1/6 (B) 1/3

(C) 1/2 (D) 2/3

Q20. The period of cos(x/6) is :

(A) 2π (B) 3π

(C) 6π (D) 12π

Note: 24-224-I-(Objective Type)- 11875 (6195)

PAPER - I (Essay Type)

Time Allowed: 2.30 hours | **Maximum Marks:** 80

Note: 224-1st Annual-(INTER PART - I) (Academic Sessions 2020 - 2022 to 2023 - 2025)

Q2. (Marks 16) Write short answers to any EIGHT (8) questions :

(i) Write the symmetric property and transitive property of equality of the real numbers.

(ii) Show that $z\bar{z} = |z|^2 \forall z \in C$

(iii) Find out real and imaginary parts of $(\sqrt{3} + i)^3$

(iv) Find the modulus of $1 - i\sqrt{3}$

(v) Construct truth table for $(p \wedge \sim p) \rightarrow q$

(vi) If a, b are elements of a group G, then show that $(ab)^{-1} = b^{-1}a^{-1}$

(vii) If $A = [1\ 2; a\ b]$ and $A^2 = [0\ 0; 0\ 0]$, find the values of a and b.

(viii) If A and B are square matrices of the same order, then explain why in general $(A - B)^2 \neq A^2 - 2AB + B^2$.

(ix) Define skew-hermitain matrix.

(x) Evaluate $\omega^{28} + \omega^{29} + 1$

(xi) When $x^4 + 2x^3 + kx^2 + 3$ is divided by $x - 2$, the remainder is 1. Find the value of k.

(xii) If α, β are the roots of $x^2 - px - p - c = 0$, prove that $(1+\alpha)(1+\beta) = 1 - c$

Q3. (Marks 16) Write short answers to any EIGHT (8) questions :

(i) Define partial fractions.

(ii) If $(7x+25)/((x+3)(x+4)) = 4/(x+3) + B/(x+4)$, then find B.

(iii) Find the number of terms in A.P if $a_1 = 3$; $d = 7$ and $a_n = 59$

(iv) If 1/a, 1/b, 1/c are in G.P., show that common ratio is $\pm\sqrt{(a/c)}$

(v) Find the sum of $9/4 + 3/2 + 1 + 2/3 + \text{-----}\infty$

(vi) If 5 is H.M. between 2 and b, then find b.

(vii) Write $(n+1)(n)(n-1)/3.2.1$ in factorial form.

(viii) Prove that ${}^nP_r = n.\ {}^{n-1}P_{r-1}$

(ix) Determine probability of getting 2 heads in two successive tosses of balanced coin.

(x) Show that $8.10^n - 2$ is divisible by 6 for $n = 1$ and $n = 2$

(xi) Find the 6th term in the expansion of $(x^2 - 3/2x)^{10}$

(xii) Using binomial theorem, find value of $\sqrt[3]{65}$ correct to three places of decimal.

Q4. (Marks 18) Write short answers to any NINE (9) questions :

(i) Verify $\tan 2\theta = 2\tan\theta/(1 - \tan^2\theta)$ for $\theta = 45°$

(ii) Prove the identity $(1 + \cos\theta)/(1 - \cos\theta) = (\operatorname{cosec}\theta + \cot\theta)^2$

(iii) If α, β and γ are the angles of triangle ABC then prove that $\tan(\alpha+\beta) - \tan\gamma = 0$

(iv) Express as product $\cos 6\theta + \cos 3\theta$

(v) Prove that $1 + \tan\alpha \tan 2\alpha = \sec 2\alpha$

(vi) Prove that period of cosine is 2π

(vii) Find the period of cosec 10x

(viii) Draw the graph of the function $y = \cos x$, $x \in [-\pi/2, \pi/2]$

(ix) Write formula for $\cos(\alpha/2)$ and $\cos(\gamma/2)$

(x) Measure of two sides of a triangle are in the ratio 3 : 2 and angle including these sides is 57°. Find the remaining two angles.

(xi) Define circum centre.

(xii) Without using calculator / table, show that $2\cos^{-1}(4/5) = \sin^{-1}(24/25)$

(xiii) Solve the trigonometric equation $\text{cosec}^2\theta = 4/3$

QSECTION - II. Note : Attempt any THREE questions.

Q5. (a) Show that the determinant $|a+\lambda\ b\ c;\ a\ b+\lambda\ c;\ a\ b\ c+\lambda| = \lambda^2(a+b+c+\lambda)$ [5] (b) If the roots of the equation $x^2 - px + q = 0$ differ by unity, prove that $p^2 = 4q + 1$ [5]

Q6. (a) Resolve $1/((x-3)^2(x+1))$ into partial fractions [5] (b) Find n so that $(a^n + b^n)/(a^{n-1} + b^{n-1})$ may be the A.M. between a and b [5]

Q7. (a) Two dice are thrown. E_1 is the event that the sum of their dots is an odd numbers and E_2 is the event that 1 is the dot on the top of the first die. Show that $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$ [5] (b) If $y = 1/3 + (1.3/2!)(1/3)^2 + (1.3.5/3!)(1/3)^3 + - - -$ prove that $y^2 + 2y - 2 = 0$ [5]

Q8. (a) Reduce $\sin^4\theta$ to an expression involving only function of multiple of θ, raised to the first power. [5] (b) Prove that $\Delta = r^2 \cot(\alpha/2) \cot(\beta/2) \cot(\gamma/2)$ [5]

Q9. (a) Find the values of all the trigonometric functions of the angle -675°. [5] (b) Prove that $\sin^{-1}(5/13) + \sin^{-1}(7/25) = \cos^{-1}(253/325)$ [5]

> Note: 24-224-I-(Essay Type)-47500

GROUP – II

Q.PAPER - I (Objective Type)

Time Allowed: 30 Minutes	**Maximum Marks:** 20	**Paper Code:** 6196

> Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. If A is a matrix of order 2 × 3, then order of A^tA is :

(A) 3 × 3 (B) 2 × 3

(C) 3 × 2 (D) 2 × 2

Q2. The equation $x(x-1) = x^2 - x$ is :

(A) Conditional (B) Identity

(C) Exponential (D) Radical

Q3. The multiplicative inverse of - i is :

(A) (1 , - 1)
(B) (0 , - 1)
(C) (0 , 1)
(D) (1 , 0)

Q4. If ω is a cube root of unity, then $(1+\omega+\omega^2)^8$ = :

(A) 0
(B) 256
(C) 256ω
(D) $256\omega^2$

Q5. Which of the following sets has closure property w.r.t. addition :

(A) { 1 }
(B) { 0 }
(C) { 0 , 1 }
(D) { 1 , - 1 }

Q6. If |A| = 9 , then $|A^t|$ is :

(A) 81
(B) 1/9
(C) - 9
(D) 9

Q7. The converse of $p \rightarrow q$ is :

(A) $\sim p \rightarrow \sim q$
(B) $\sim q \rightarrow p$
(C) $q \rightarrow p$
(D) $p \rightarrow \sim q$

Q8. If A = {}, then the power set of A is :

(A) φ
(B) {0}
(C) {}
(D) {φ}

Q9. If $4^{1+x} = 2$, then x = :

(A) 0
(B) - 2
(C) - 1/2
(D) 1/2

Q10. If $A \cap B = A$, then :

(A) $B \subseteq A$
(B) $A \subseteq B$
(C) $A \cup B = A$
(D) $B \cup A = A$

Q11. $\sin(270° + \theta)$ = :

(A) sin θ
(B) cos θ
(C) -cos θ
(D) -sin θ

Q12. Which cannot be the term of a G.P. :

(A) 1
(B) - 1
(C) 0
(D) i

Q13. If sin x = -√3/2 , then the reference angle is :

(A) -π/6 (B) π/6

(C) -π/3 (D) π/3

Q14. Which angle is quadrantal angle :

(A) 45° (B) 60°

(C) 120° (D) 270°

Q15. With usual notation, abc/4R = :

(A) r (B) r_1

(C) Δ (D) r_2

Q16. H.M. between 3 and 7 is :

(A) 5 (B) √21

(C) ±√21 (D) 21/5

Q17. The number of terms in the expansion of $(a+x)^n$ is :

(A) n - 1 (B) n

(C) n + 2 (D) n + 1

Q18. The period of cos 2x is :

(A) π (B) 2π

(C) 4π (D) π/2

Q19. If r = n , then ${}^nC_r$ = :

(A) 0 (B) 1

(C) n (D) n!

Q20. $\sin^{-1}(0) + \cos^{-1}(0)$ = :

(A) 0 (B) π/2

(C) π/3 (D) π/4

Note: 25-224-II-(Objective Type)- 11750 (6196)

PAPER - I (Essay Type)

Time Allowed: 2.30 hours | **Maximum Marks:** 80

Note: 224-1st Annual-(INTER PART - I) (Academic Sessions 2020 - 2022 to 2023 - 2025)

Q2. (Marks 16) Write short answers to any EIGHT (8) questions :

(i) Show that $z^2\bar{z}^2$ is a real number.

(ii) Find the modulus of $1 - i\sqrt{3}$

(iii) Simplify by justifying each step $(1/4 + 1/5)/(1/4 - 1/5)$

(iv) Check the closure property w.r.t. addition and multiplication for the set { 0 , -1 }

(v) Determine whether the statement $p \wedge \sim p$ is tautology or not.

(vi) Define semi-group.

(vii) If A = [1; 1+i; i], find $A(\bar{A})^t$

(viii) Define reduced echelon form of a matrix, with example.

(ix) If A = [2 -1; 3 1], verify that $(A^t)^{-1} = (A^{-1})^t$

(x) Discuss nature of roots of $9x^2 - 12x + 4 = 0$

(xi) Solve the equations $x^2 + y^2 = 25$, $2x^2 + 3y^2 = 6$

(xii) Find the condition that one root of $x^2 + px + q = 0$ is square of other.

Q3. (Marks 16) Write short answers to any EIGHT (8) questions :

(i) Define proper rational fraction.

(ii) For the identity $1/((x-1)(2x-1)(3x-1)) = A/(x-1) + B/(2x-1) + C/(3x-1)$ calculate the value of A.

(iii) Find the next two terms of 1, 3, 7, 15, 31, ----

(iv) How many terms are there in the A.P. in which $a_1 = 11$, $a_n = 68$, $d = 3$

(v) Find three A.Ms between $\sqrt{2}$ and $3\sqrt{2}$.

(vi) Find the 12th term of 1 + i, 2i, - 2 + 2i, -----

(vii) Show that ${}^{16}C_{11} + {}^{16}C_{10} = {}^{17}C_{11}$

(viii) Evaluate ${}^{12}C_3$

(ix) What is sample space and events?

(x) State principle of mathematical induction.

(xi) Calculate $(9.98)^4$ by means of binomial theorem.

(xii) Prove that $n! > 2^n - 1$ for n = 4,5

Q4. (Marks 18) Write short answers to any NINE (9) questions :

(i) What is length of an arc intercepted on a circle of radius 14 cm by the arms of a central angle 45°?

(ii) Convert 54°45' into radians.

Note: (Turn Over)

(iii) If α, β, γ are angles of triangle ABC then prove that $\cos((\alpha+\beta)/2) = \sin(\gamma/2)$

(iv) Find the value of $\cos(\pi/12)$

(v) Express $\sin(x+30°) + \sin(x-30°)$ as a product.

(vi) Define periodic function and period of trigonometric function.

(vii) Find period of $\cos(x/6)$

(viii) Draw the graph of $y = \sin x$ from 0 to π.

(ix) State law of sines.

(x) If sides of triangle are 16, 20, 23, find its greatest angle.

(xi) Show that $r_1 = s \tan(\alpha/2)$

(xii) Find value of $\cos(\sin^{-1}(1/\sqrt{2}))$

(xiii) Show that $\tan(\sin^{-1}x) = x/\sqrt{(1-x^2)}$

QSECTION - II. Note : Attempt any THREE questions.

Q5. (a) Solve the system of equations by Cramer's rule : $2x + 2y + z = 3$; $3x - 2y - 2z = 1$; $5x + y - 3z = 2$ [5] (b) If α, β roots of $x^2 - 3x + 5 = 0$ form the equation whose roots are $(1-\alpha)/(1+\alpha)$ and $(1-\beta)/(1+\beta)$ [5]

Q6. (a) Resolve $x^4/(1-x^4)$ into partial fractions [5] (b) The sum of an infinite geometric series is 9 and the sum of the squares of its terms is 81/5. Find the series. [5]

Q7. (a) Find the values of n and r when ${}^{n-1}C_{r-1} : {}^{n}C_{r} : {}^{n+1}C_{r+1} = 3:6:11$ [5] (b) If x is so small that its cube and higher powers can be neglected, then show that : $\sqrt{((1-x)/(1+x))} \approx 1 - x + x^2/2$ [5]

Q8. (a) Reduce $\cos^4\theta$ to an expression involving only function of multiples of θ, raised to the first power. [5] (b) Prove that $r_3 = 4R \cos(\alpha/2) \cos(\beta/2) \sin(\gamma/2)$ [5]

Q9. (a) Show that the area of a sector of a circular region of radius r is $(1/2)r^2\theta$, where θ is the circular measure of the central angle of the sector. [5] (b) Prove that $\sin^{-1}(1/\sqrt{5}) + \cot^{-1}3 = \pi/4$ [5]

Note: 25-224-II-(Essay Type)-47000